02 · User Personas & Scenarios

> **Assignment:** *"User Personas and Scenarios — describe the different users that will interact with the CDSS, and provide 2 user scenarios that illustrate examples of how the user will interact with the system."*

2.1 Overview of the user groups

WundCheck has three user groups at three levels of the care system. Only one of them actively operates the app — the other two consume its output.

Persona	Role	Interaction	Level
Peter Brunner	Patient	Records data daily, receives a recommendation	Primary user
Sandra Meier	Medical practice assistant / nurse	Receives alerts, triages, organises appointments	Secondary user
Dr. Anna Keller	Surgeon	Reads the course in the hospital information system, uses aggregated data	Tertiary user

2.2 Persona 1 — Peter Brunner, patient *(primary user)*

Profile	58, commercial employee, total knee replacement 4 days ago (implant surgery), lives with his wife, little medical background knowledge, wears reading glasses, uses his smartphone for everyday apps
Goals	To be sure that healing is progressing normally · not to call the hospital about "trivialities" · not to jeopardise the suture removal appointment
Frustrations	Does not know what "normal" looks like after surgery · finds the hospital's brochure too text-heavy · afraid of overlooking warning signs (his brother had a wound infection)
Tech literacy	Medium — WhatsApp, online banking, but no willingness to install an app or create an account
Relation to the CDSS	Completes the daily check in the evening, selects the wound appearance via a graphic, follows the traffic-light recommendation
Success criterion from his point of view	"In the evening I know whether everything is fine — without having to call anyone."

→ Design consequences: Images instead of technical terms (NFR-5) · no login, no app store (NFR-1) · reassuring feedback for normal findings is just as important as the warning (FR-4) · large type, high contrast.

2.3 Persona 2 — Sandra Meier, medical practice assistant *(secondary user)*

Profile	34, medical practice assistant in the orthopaedic outpatient clinic, triages telephone calls and messages daily, high workload
Goals	To recognise and prioritise serious cases quickly · to reduce unnecessary calls · clear, structured information instead of vague patient descriptions

Frustrations	Telephone triage eats up time · "the wound looks odd" is not usable information · lack of documentation of the course of healing
Relation to the CDSS	Receives orange/red alerts with a structured wound score and course, triages and organises a call-back or an appointment, sees the data in the record
Success criterion from her point of view	"Within 10 seconds I can see whether this case needs an appointment today."

→ Design consequences: The alert contains the classification, the triggering rule, vital values and the 7-day course — not just "patient reports a problem" · alerts only for ORANGE/RED, otherwise alert fatigue · structured data instead of free text.

2.4 Persona 3 — Dr. med. Anna Keller, surgeon *(tertiary user)*

Profile	45, specialist in orthopaedics, operates and carries out follow-up examinations
Goals	To see the course of healing at a glance at the suture removal appointment · to detect complications early · to monitor the SSI rate of her procedures
Frustrations	The course between discharge from hospital and the follow-up examination is a black box · retrospective patient recall is unreliable
Relation to the CDSS	Reads the daily wound score course directly in the hospital information system (via the FHIR interface), uses aggregated data for quality monitoring
Success criterion from her point of view	"I see the complete course without having to open every single answer."

→ Design consequences: Additionally store the classification as a **FHIR** `Observation`, so that the course can be displayed as a curve without opening the `QuestionnaireResponse` · data must also be transmitted for GREEN, otherwise the baseline is missing.

2.5 Scenario A — Normal healing with a scab *(green path)*

Day 6 after the knee surgery.

Peter receives the daily reminder at 18:00. He measures **36.9 °C**, reports **pain 2/10** (better than yesterday) and sees a brownish scab on the wound — this worries him, his brother had "something like that too" at the time.

In the app he selects **"Calm & dry"** ("Ruhig & trocken") for the wound appearance and answers the question *"Has a scab / crust formed?" ("Hat sich Wundschorf / eine Kruste gebildet?")* with **Yes**.

WundCheck displays 🟢 **GREEN** with the note:

> "A scab is a normal sign of healing — do not scratch it off, keep the wound dry." ("Wundschorf ist ein normales Zeichen der Heilung — nicht abkratzen, Wunde trocken halten.")

Peter is reassured and does **not** call the hospital. The daily entry is transmitted to the patient record as a FHIR data set. At the suture removal appointment on day 12, Dr. Keller sees the complete, unremarkable course.

> **What this scenario shows:** the reassuring function is not a side effect but a goal in its own right. A CDSS that only warns would not have created any benefit here — the avoided unnecessary call *is* the benefit. Clinically represented in rule **I4** (scab = information, not a warning sign) and documented as challenge **C-04**.

2.6 Scenario B — Suspected infection with an alert *(orange path)*

Day 7.

Peter's wound has been distinctly more reddened since yesterday, and he measures **38.3 °C**. In the app he selects the wound appearance **"Distinctly reddened & swollen"** ("Deutlich gerötet & geschwollen"), for the erythema **"extending beyond the wound edge"** ("über den Wundrand hinaus") and for the discharge **"cloudy/yellowish"** ("trüb/gelblich").

WundCheck classifies 🟠 **ORANGE**:

> "Suspected infection — contact your practice today." ("Infektionsverdacht — kontaktieren Sie heute Ihre Praxis.")

At the same time, a structured alert goes to the practice. **Sandra** sees it at 08:10 in the dashboard: classification, triggering rule, temperature, wound appearance and the 7-day course are immediately visible — no guessing game on the telephone. She calls Peter back within 30 minutes and gives him an appointment the same afternoon.

Dr. Keller confirms a superficial wound infection and starts treatment **two days earlier** than would probably have been the case without WundCheck.

> **What this scenario shows:** the benefit does not arise in the app but at the interface. The classification alone would only have prompted Peter to call — the structured alert additionally shortens the triage time in the practice.

2.7 Scenario C — Borderline case *(stress test, not part of the compulsory submission)*

Day 2, late in the evening, no mobile network coverage in the holiday home.

Peter feels **"really ill"**, but measures only **37.4 °C**; the wound itself looks unremarkable. He completes the check offline.

WundCheck classifies 🟠 **ORANGE** via rule **R9** (severe subjective feeling of illness even without measurable fever) and points out that transmission to the practice is pending until network coverage is available again — the recommendation for action nevertheless applies immediately.

> **What this scenario shows:** rule R9 has only existed since bug **B-02** — in the first version, `wellbeing` was collected but was not used in any rule. A patient with a severe feeling of illness but without fever would have received **GREEN**. This exact pattern is dangerous in early sepsis. See 10 QA testing.

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